

Day 17: Hybrid Tracking (Motion + Color)

Outcomes:

- Understand why single-method tracking is unreliable
- Learn the concept of combining motion detection with color detection
- Know how motion limits the search area
- Understand how hybrid tracking improves accuracy and stability

The limitation of single-method tracking

So far you have used two different detection ideas:

Color-based detection

HSV → Mask → Contours → Largest contour

Problems:

- lighting changes
- background has similar colors
- shadows affect detection

Motion-based detection

Background subtraction → Foreground mask → Contours

Problems:

- detects *everything that moves*
- shadows and lighting shifts create false detections
- stationary objects disappear

Neither method is reliable alone.

The hybrid solution

Real systems combine both ideas.

The logic becomes:

Motion detection → where something changed

Color detection → what object we want

Motion finds candidate regions.

Color verifies the correct object.

How hybrid tracking works conceptually

Step 1 — Motion detection

Detect moving regions.

Step 2 — Region filtering

Limit the search to those regions only.

Step 3 — Color verification

Check if the moving region contains the target color.

Step 4 — Tracking

Apply the same pipeline you already built:

Largest contour:

→ center

→ smoothing

→ trail

→ bounding box

Why this improves reliability

Hybrid tracking reduces:

False positives from color detection: background object with same color

False positives from motion detection: moving shadow or random motion

The system now asks two questions:

Did something move?

AND

Is it the correct color?

Both conditions must be true.

Practical advantage

Hybrid tracking:

- ignores stationary background objects
- ignores random movement
- focuses detection on the actual target

This approach is used in many real-world systems such as:

- gesture recognition
- object tracking
- surveillance motion filters